```python
import pandas as pd
```

```python
df = pd.read_csv("additinal columns.csv")
```

```python
df.head()
```

tomer Name	Customer Age	Customer Gender	Product Purchased	Date of Purchase	Ticket Type	Ticket Subject	Ticket Status	Ticket Priority	Ticket Channel	Custon Satisfacti Rati
lelinda Smith	18	Other	Microsoft Office	2020-04-04	Cancellation request	Battery life	Closed	High	Chat	
Lisa Morales	18	Male	Canon EOS	2020-06-12	Technical issue	Hardware issue	Open	Low	Chat	N
ennifer Lucas	18	Female	LG Smart TV	2020-11-22	Technical issue	Peripheral compatibility	Closed	Low	Social media	
Allison ohnson	18	Female	Sony PlayStation	2020-05-06	Technical issue	Network problem	Open	High	Chat	N
Rachael Walker	18	Male	Sony 4K HDR TV	2020-01-06	Refund request	Product recommendation	Open	Medium	Social media	N

```python
df.columns = df.columns.str.strip()
```

```python
df = df.sort_values(by='Ticket ID', ascending=True)
```

```python
df = df.reset_index(drop=True)
```

```python
df.to_csv("additinal columns.csv", index=False)
```

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```
[6]: df = df.reset_index(drop=True)
```

```
[8]: df.to_csv("additinal columns.csv", index=False)
```

```
[9]: df['First_Response_Minutes'] = df['First_Response_Minutes'].round(0)
```

```
[10]: df.to_csv("additinal columns.csv", index=False)
```

```
[11]: df['First_Response_Minutes'] = df['First_Response_Minutes'].astype(int)
```

```
[12]: df.to_csv("additinal columns.csv", index=False)
```

```
[13]: df['First_Response_Minutes'] = df['First_Response_Minutes'].fillna(0).astype(int)
```

```
[14]: df.to_csv("additinal columns.csv", index=False)
```

```
[15]: df['First_Response_Minutes'] = df['First_Response_Minutes'].astype(int)
```

```
[16]: df['First_Response_Minutes'] = df['First_Response_Minutes'].fillna(0).astype(int)
```

```
[17]: import numpy as np
      df['First_Response_Minutes'] = np.floor(df['First_Response_Minutes'])
```

```
[18]: df['First_Response_Minutes'] = df['First_Response_Minutes'].round().astype(int)
```

```
[19]: df.to_csv("additinal columns.csv", index=False)
```

```
[20]: df['Customer Satisfaction Rating'] = df['Customer Satisfaction Rating'].fillna(0)
```

```
[21]: df['First_Response_Minutes'] = df['First_Response_Minutes'].round().astype(int)
```

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[22]:
```
df['Average_First_Response_Minutes'] = (
    df['Average_First_Response_Minutes']
    .round()
    .astype(int)
)
```

[23]:
```
df.head()
```

[23]:

	Ticket ID	Customer Name	Customer Age	Customer Gender	Product Purchased	Date of Purchase	Ticket Type	Ticket Subject	Ticket Status	Ticket Priority	Ticke Channe
0	1	Marisa Obrien	32	Other	GoPro Hero	2021-03-22	Technical issue	Product setup	Pending Customer Response	Critical	Socia medi
1	2	Jessica Rios	42	Female	LG Smart TV	2021-05-22	Technical issue	Peripheral compatibility	Pending Customer Response	Critical	Cha
2	3	Christopher Robbins	48	Other	Dell XPS	2020-07-14	Technical issue	Network problem	Closed	Low	Socia medi
3	4	Christina Dillon	27	Female	Microsoft Office	2020-11-13	Billing inquiry	Account access	Closed	Low	Socia medi
4	5	Alexander Carroll	67	Female	Autodesk AutoCAD	2020-02-04	Billing inquiry	Data loss	Closed	Low	Ema

[24]:
```
df.to_csv("supportlytics_cleaned_dataset.csv", index=False)
```

[29]:
```
df['Priority_Avg_Response'] = df['Priority_Avg_Response'].round().astype(int)
```

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[29]:
```
df['Priority_Avg_Response'] = df['Priority_Avg_Response'].round().astype(int)
```

[30]:
```
df['Priority_Avg_Response'] = df['Priority_Avg_Response'].astype(int)
```

[31]:
```
df['Priority_Avg_Response'] = df['Priority_Avg_Response'].astype(int)
```

[32]:
```
print(df.columns)
```

```
Index(['Ticket ID', 'Customer Name', 'Customer Age', 'Customer Gender',
       'Product Purchased', 'Date of Purchase', 'Ticket Type',
       'Ticket Subject', 'Ticket Status', 'Ticket Priority', 'Ticket Channel',
       'Customer Satisfaction Rating', 'First_Response_Minutes',
       'First_Response_Date', 'Average_First_Response_Minutes',
       'Priority_Avg_Response'],
      dtype='object')
```

[33]:
```
df['Priority_Avg_Response'] = (
    df['Priority_Avg_Response']
    .round()
    .astype(int)
)
```

[34]:
```
df[['Priority_Avg_Response']].head()
```

[34]:

	Priority_Avg_Response
0	20
1	20
2	20
3	20

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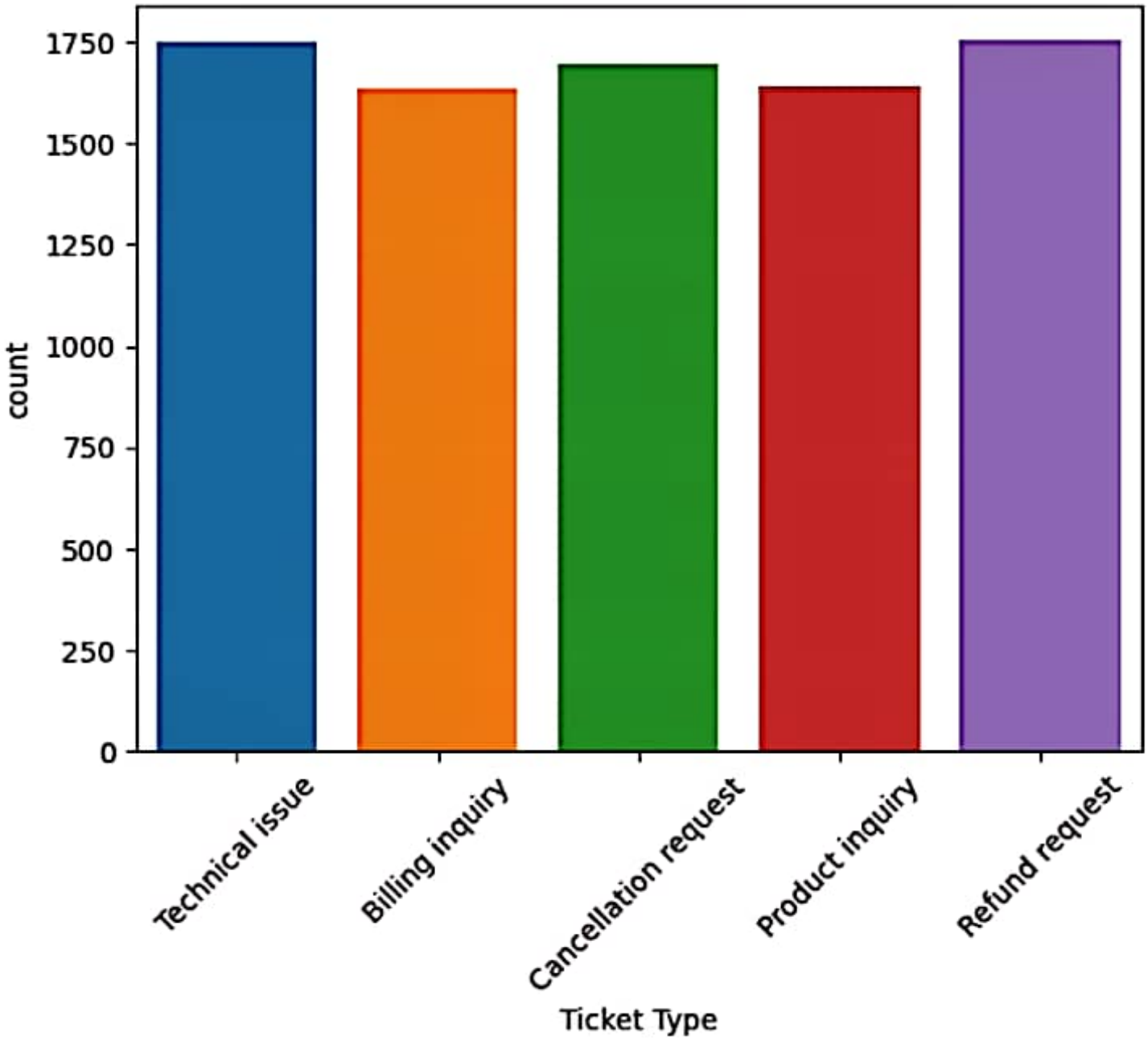
[41]:
```
sns.countplot(x='Ticket Type', data=df)
plt.xticks(rotation=45)
plt.show()
```



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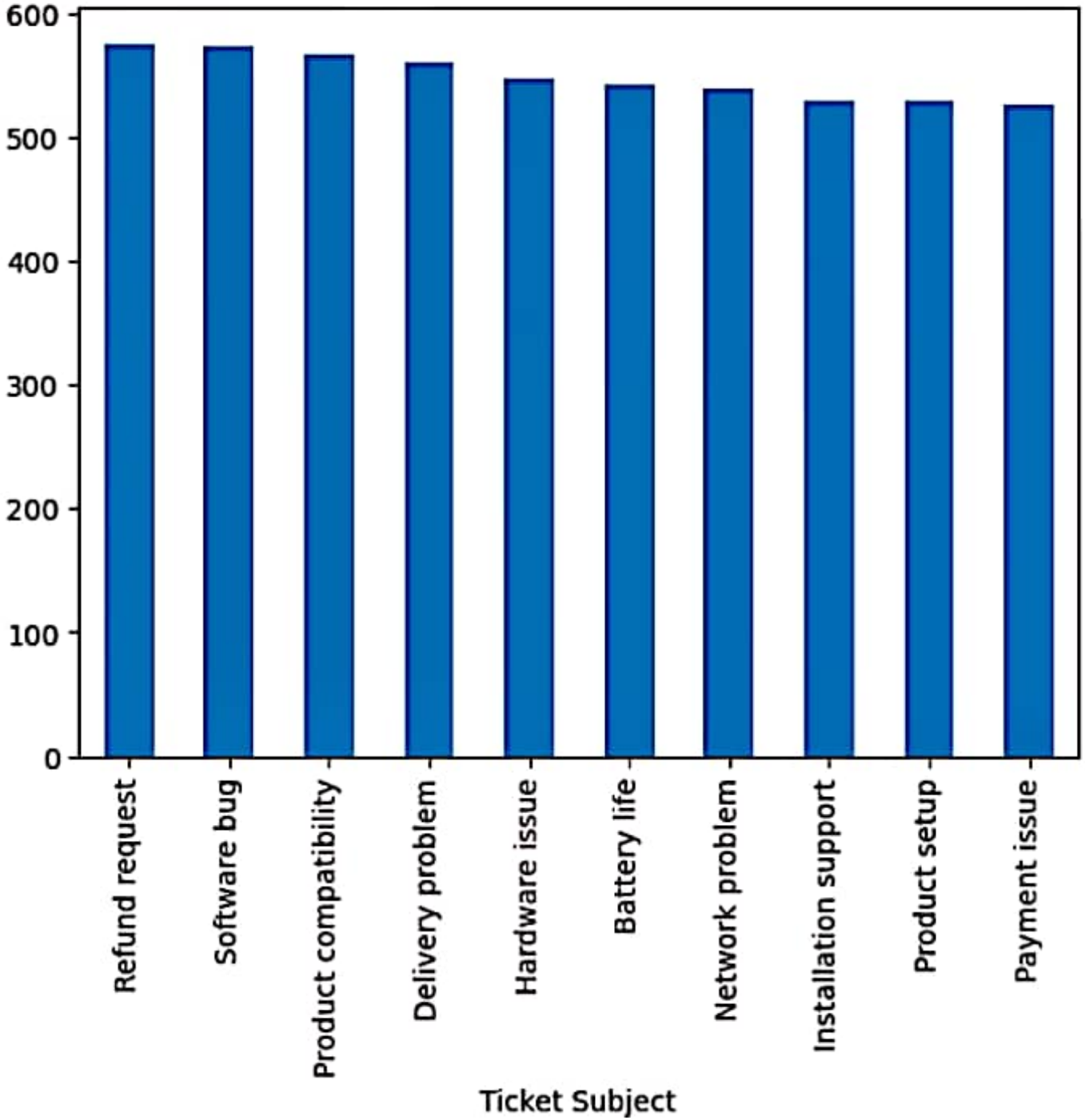
[44]: <Axes: xlabel='Ticket Subject'>



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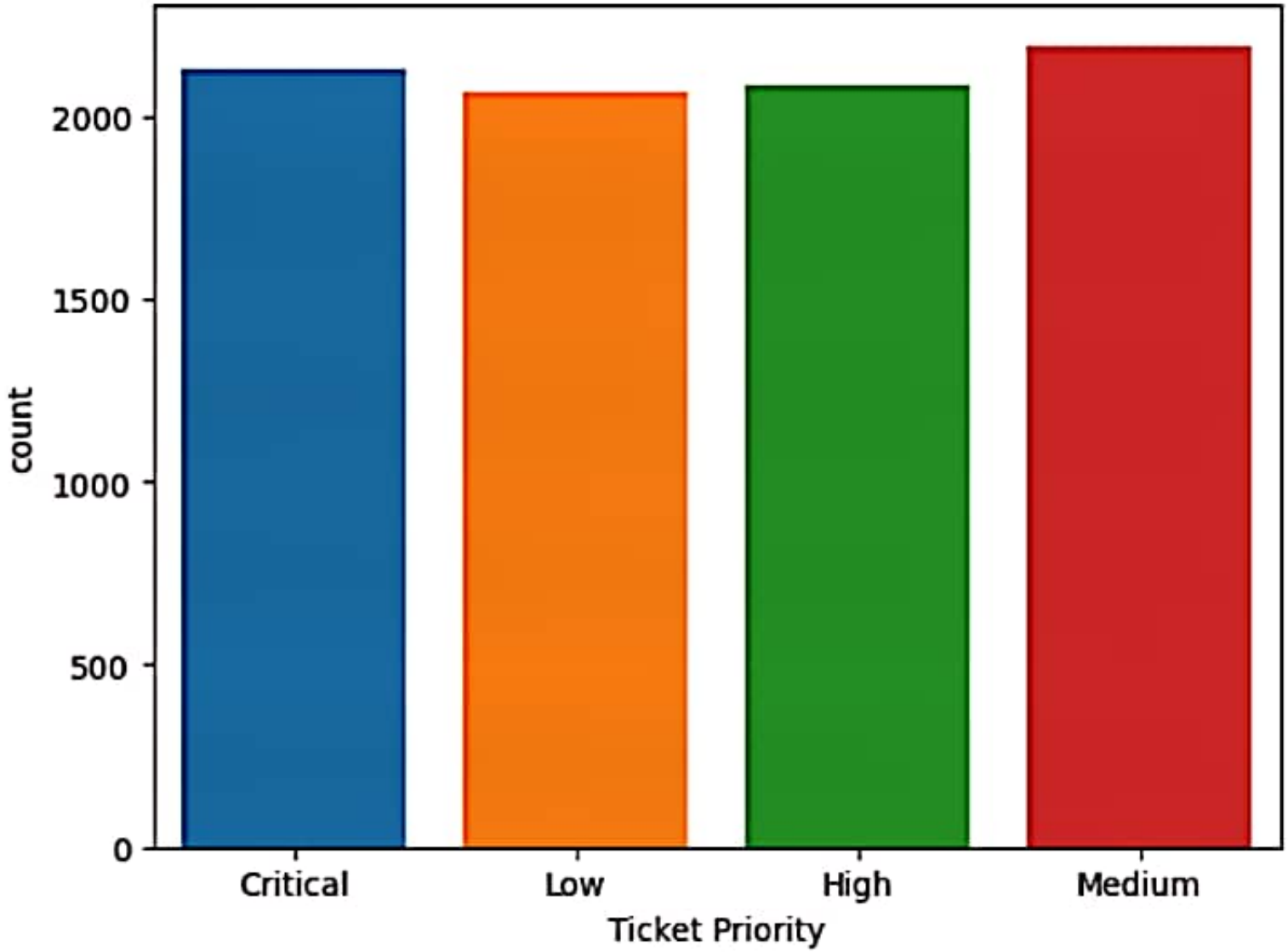
Ticket Subject

[45]:
```
sns.countplot(x='Ticket Priority', data=df)
```

[45]: <Axes: xlabel='Ticket Priority', ylabel='count'>



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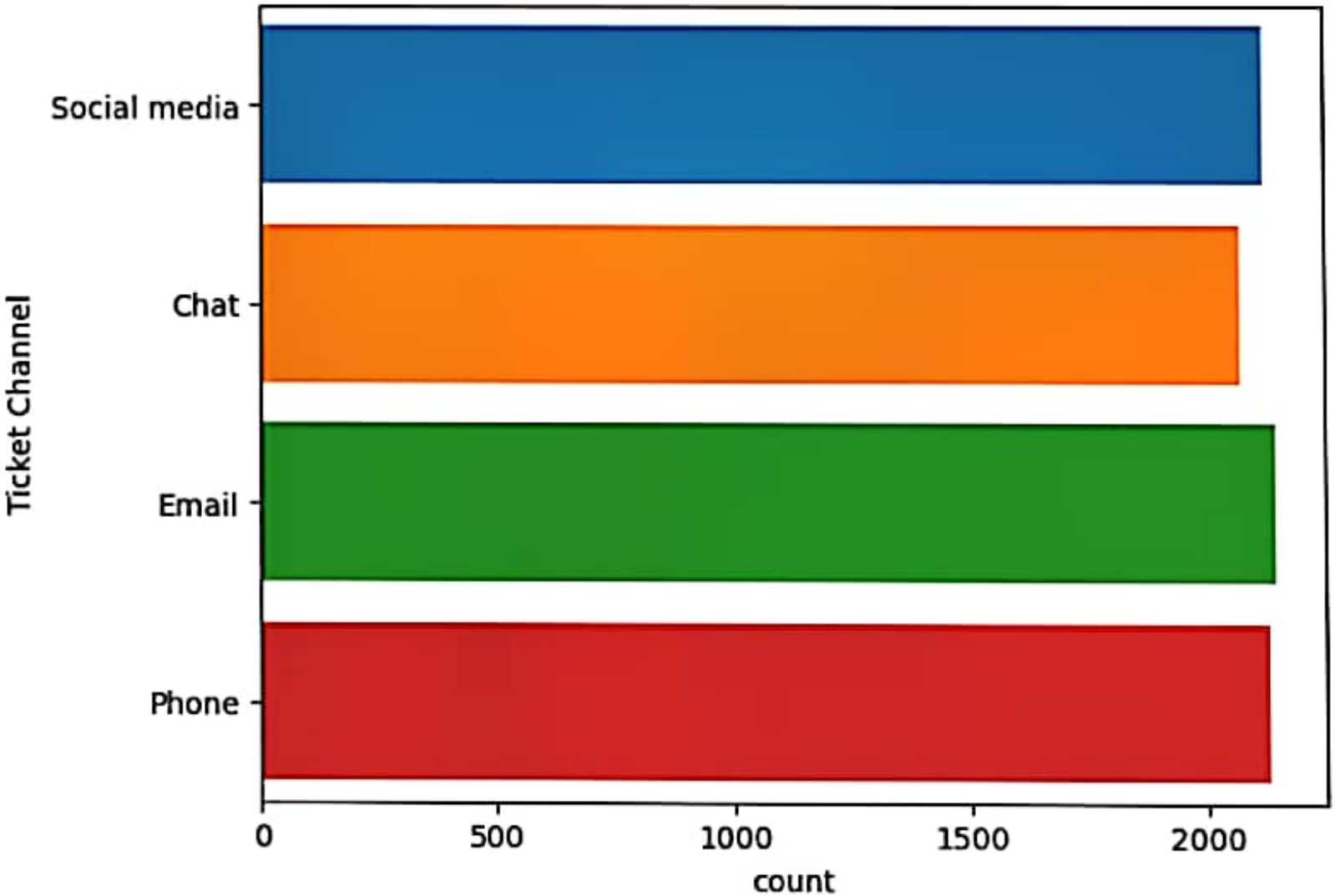
[46]:
```
sns.countplot(y='Ticket Channel', data=df)
```

[46]: <Axes: xlabel='count', ylabel='Ticket Channel'>



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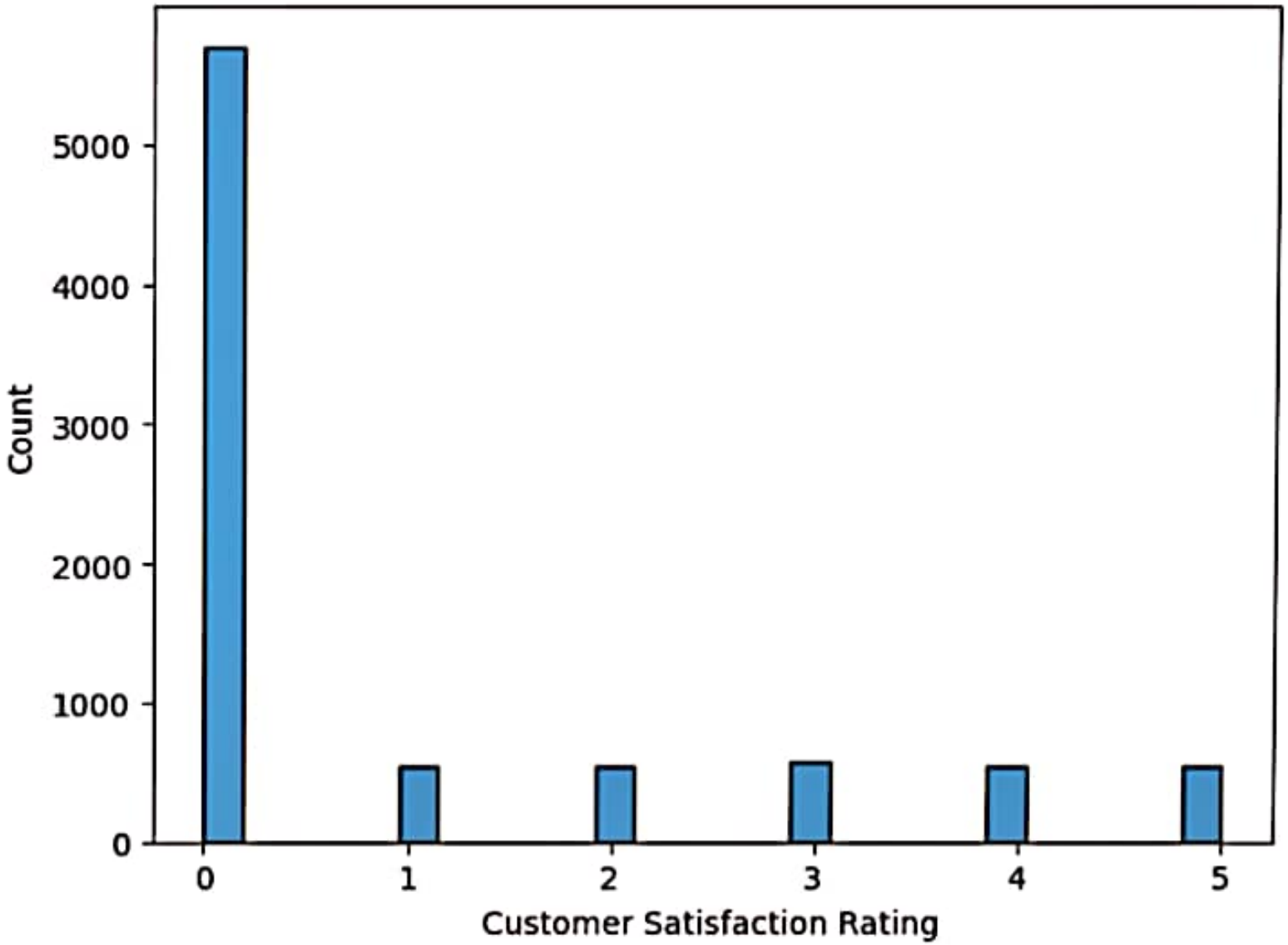
[48]:
```
sns.histplot(df['Customer Satisfaction Rating'])
plt.show()
```



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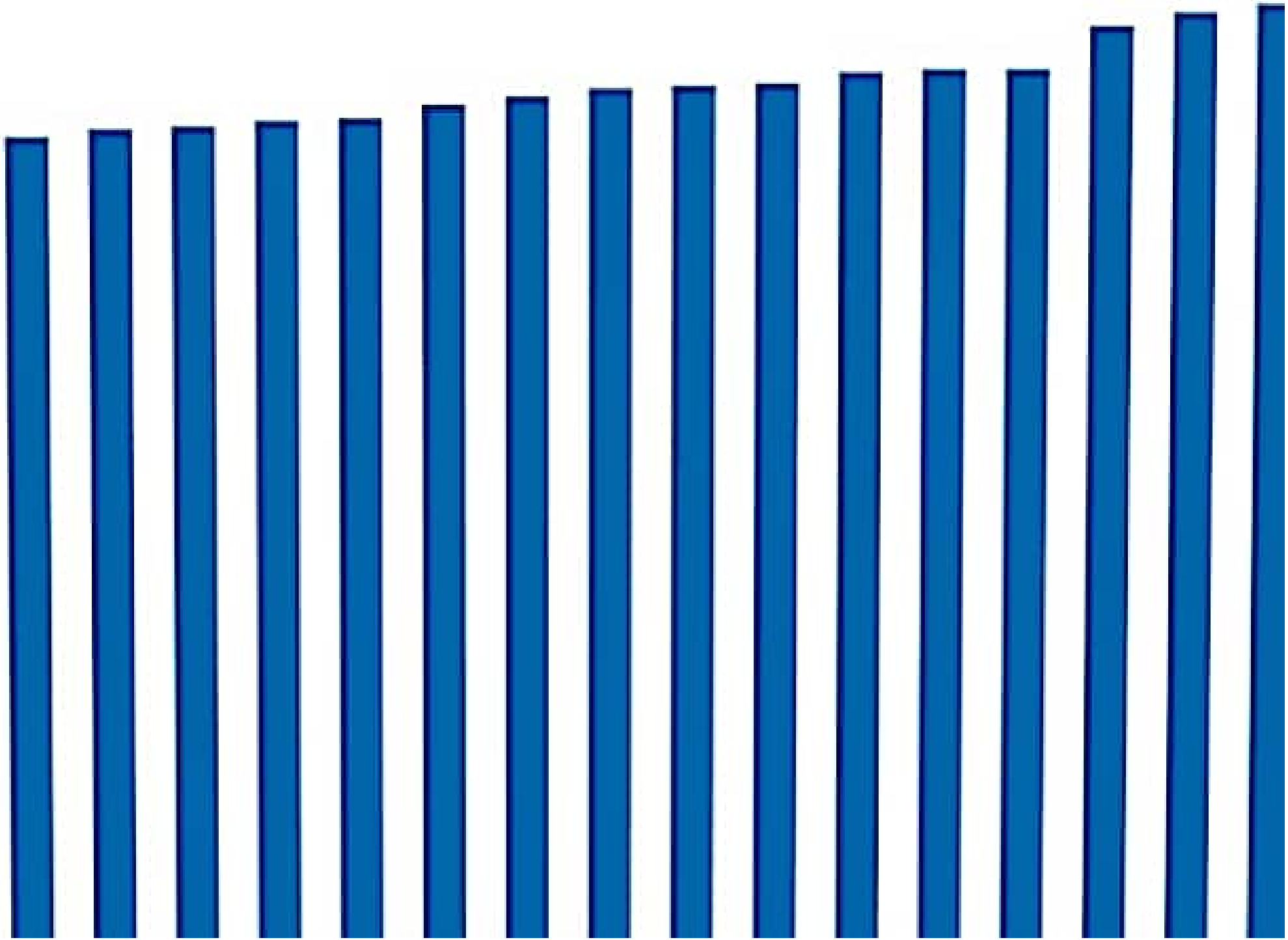

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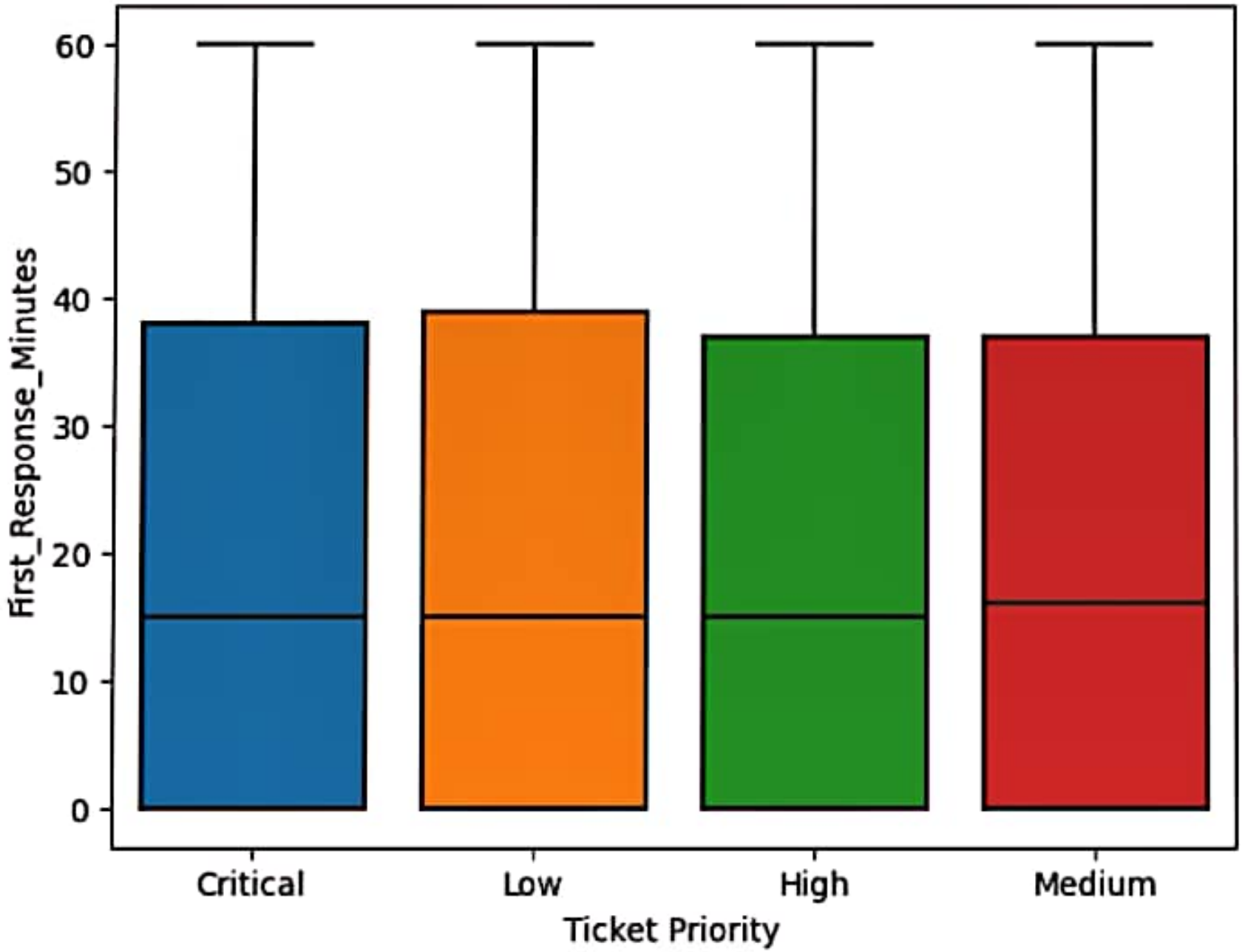
[50]:
```
sns.boxplot(x='Ticket Priority', y='First_Response_Minutes', data=df)
plt.show()
```



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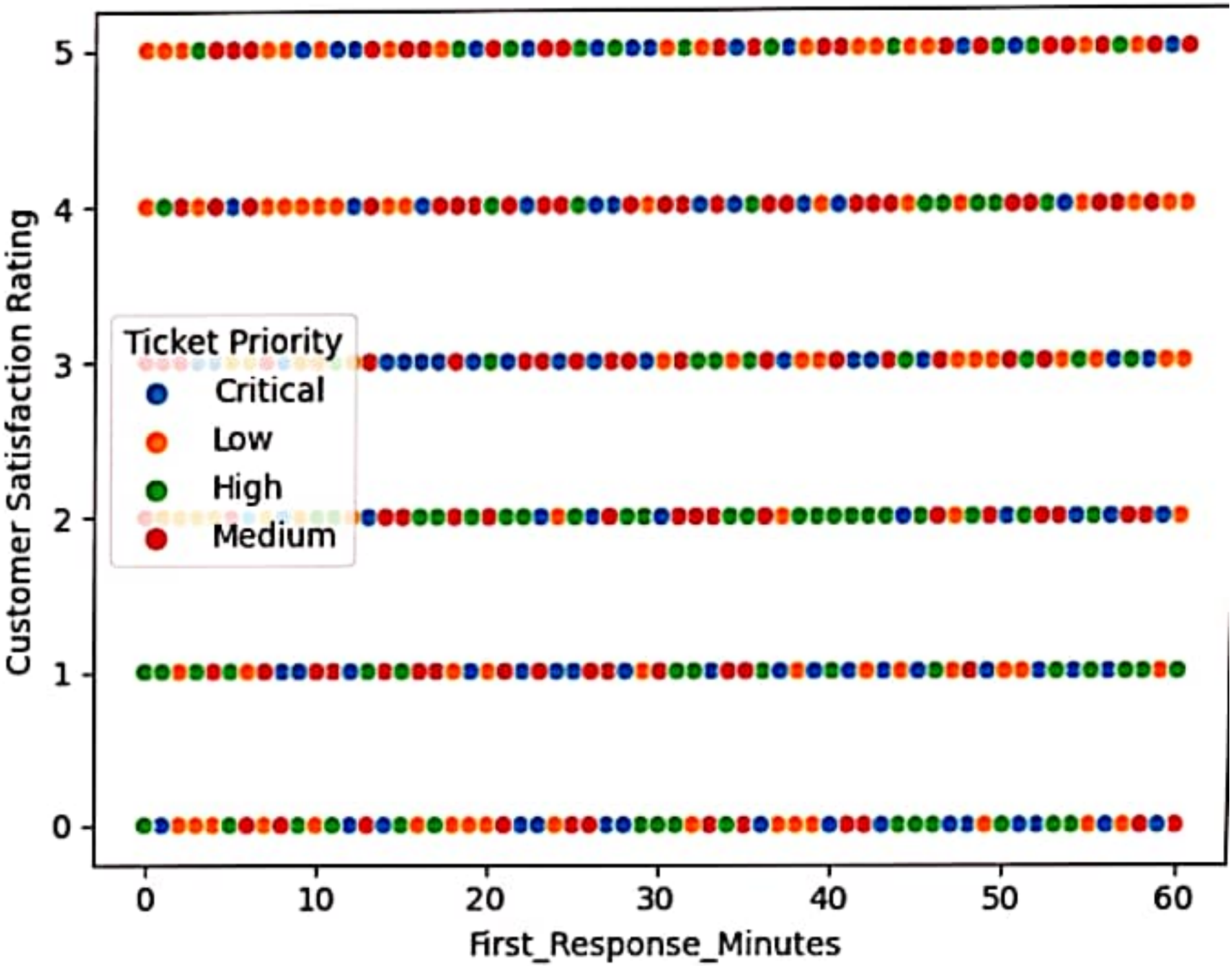

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[4]:
```python
import pandas as pd

df = pd.read_csv("final_data_set.csv")
```

[5]:
```python
df.head()
```

[5]:

ket nel	Customer Satisfaction Rating	First_Response_Minutes	First_Response_Date	Average_First_Response_Minutes	Priority_Avg_Response
cial dia	0.0	16	2023-06-01	30	20
hat	0.0	46	2023-06-01	30	20
cial dia	3.0	15	2023-06-01	30	20
cial dia	3.0	30	2023-06-01	30	20
nail	1.0	13	2023-06-01	30	20

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[6]:
```
df['Cluster'] = df['Ticket Subject']
```

[7]:
```
df.head()
```

[7]:

omer ction ating	First_Response_Minutes	First_Response_Date	Average_First_Response_Minutes	Priority_Avg_Response	Cluste
0.0	16	2023-06-01	30	20	Produc setu
0.0	46	2023-06-01	30	20	Periphera compatibilit
3.0	15	2023-06-01	30	20	Networ probler
3.0	30	2023-06-01	30	20	Accour acces
1.0	13	2023-06-01	30	20	Data los

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[8]:
```
df[['Ticket Subject', 'Cluster']].head()
```

[8]:

	Ticket Subject	Cluster
0	Product setup	Product setup
1	Peripheral compatibility	Peripheral compatibility
2	Network problem	Network problem
3	Account access	Account access
4	Data loss	Data loss

[9]:
```
df['Cluster'].value_counts().head(10)
```

[9]:
```
Cluster
Refund request           576
Software bug             574
Product compatibility    567
Delivery problem         561
Hardware issue           547
Battery life             542
Network problem          539
Installation support     530
Product setup            529
Payment issue            526
Name: count, dtype: int64
```

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[10]:
```
df.groupby('Cluster')['First_Response_Minutes'].mean().sort_values(ascending=False).head(10)
```

[10]:
```
Cluster
Product recommendation    22.001934
Account access            21.779961
Product setup             21.465028
Hardware issue            20.480804
Refund request            20.465278
Network problem           20.441558
Battery life              20.204797
Payment issue             20.115970
Delivery problem          20.081996
Installation support      19.896226
Name: First_Response_Minutes, dtype: float64
```

[11]:
```
df['Cluster'].value_counts().head(10)
```

[11]:
```
Cluster
Refund request           576
Software bug             574
Product compatibility    567
Delivery problem         561
Hardware issue           547
Battery life             542
Network problem          539
Installation support     530
Product setup            529
Payment issue            526
Name: count, dtype: int64
```

[12]:
```
df.groupby('Cluster')['First_Response_Minutes'] \
  .mean().sort_values(ascending=False).head(10)
```

[12]:
```
Cluster
```

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[13]:
```
df.groupby('Cluster')['Customer Satisfaction Rating'] \
  .mean().sort_values().head(10)
```

[13]:
```
Cluster
Data loss                    0.887984
Delivery problem             0.889483
Payment issue                0.906844
Cancellation request         0.932238
Battery life                 0.942804
Peripheral compatibility     0.943548
Refund request               0.947917
Installation support         0.949057
Account access               0.988212
Display issue                0.997908
Name: Customer Satisfaction Rating, dtype: float64
```

[14]:
```
df.to_csv("final_dataset_with_cluster.csv", index=False)
```

[15]:
```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

# display settings
plt.style.use('default')
sns.set()
```

[16]:
```
df = pd.read_csv("final_dataset_with_cluster.csv")
df.head()
```

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